Rudder Pedal Assembly – Maintenance Practices

1. Removal/Installation

Refer to Figure 1 for details.

A. Remove Rudder Pedal Assembly

1) Remove clevis pins from wheel brake lever rods, located at bulkhead station 60.00.

2) Disconnect left and right push rods from rudder torque tube bellcrank.

3) Disconnect pedal adjustment catch assembly at lower bolt.

4) Support rudder pedal assembly and remove bolts securing support pivots to bulkhead brackets.

5) Remove rudder pedal assembly from aircraft.

B. Install Rudder Pedal Assembly

1) Position rudder pedal assembly in aircraft and support.

2) Install bolts, spacers, washers and nuts to secure support pivots to bulkhead brackets. Position additional spacing washers as required to obtain maximum end play of 0.03 inch. Tighten nuts and secure with cotter pins.

3) Connect pedal adjustment catch assembly lower end using bolt, washer and nut. Catch assembly must be free to rotate on bolt. Secure nut with cotter pin.

4) Using clevis and cotter pins connect wheel brake lever rods to levers at bulkhead station 60.00.

5) Position rudder pedals to obtain a dimension of 5.4 ± 0.125 inches between bulkhead at station 60.00 to center of pedal pivots, with adjustment catch in upper hole.

6) With rudder set to neutral and pedals set as in Para 1.B.5), connect left and right push rods to rudder torque tube bellcrank. Position washer between rod and bellcrank to ensure clearance of parking brake tube.

7) Rig rudder control system (refer to 27-20-00, Rudder Control System – General Data).

8) Adjust wheel brake rods forward of rudder pedals to obtain pedal angle of 73° in the neutral position.

Note Angle measured between a line taken from heel and toe rubbers and flight compartment floor.

9) Check wheel brake rigging (refer to 32-40-00, Wheels and Brakes (Mod 6/2002) – General Data).

2. Inspection/Check

A. Wear Check – Rudder Push Rod and Wheel Brake Rod Bearings

Refer to Figure 1 for details.

1) Gain access to pilot's LH and RH rudder push rods.

2) Remove cotter pins, nuts, washers and bolts attaching upper end of rudder push rods to rudder pedal support assembly.

3) Check for bearing wear at lower end of rudder push rods, as indicated by play between ball and housing.

4) If evidence of wear exists, omit Para 2.A.5) through Para 2.A.10) and proceed as follows:

a) Remove push rods from aircraft.

b) Check that play between ball and housing, in upper and lower bearings, does not exceed 0.020–inch and that radial play does not exceed 0.006–inch.

c) Install serviceable pushrods in aircraft. Connect with bolts, washers and nuts and lock with cotter pins.

Note Washer to be positioned between push rod and rudder torque tube bellcrank, as required, to ensure push rod clearance of parking brake tube.

5) If no evidence of wear exists, connect upper end of push rods to rudder pedal support assembly, using bolts, washers and nuts. Position one washer between support assembly and each rod end and remaining washer(s) under nut.

Note Do not lock with cotter pins at this stage.

6) Remove cotter pins, nuts, washers and bolts attaching lower end of rudder push rods to rudder torque tube bellcrank.

7) Check for bearing wear at upper end of rudder push rods (refer to Para 2.A.3)).

8) If evidence of wear exists, proceed as follows:

a) Remove push rods from aircraft.

b) Check that play between ball and housing does not exceed 0.020–inch, and radial play does not exceed 0.006–inch.

9) If no evidence of wear exists, connect lower end of push rods to rudder pedal support assembly, using bolts, washers and nuts.

Note Washer to be positioned between push rod and rudder torque tube bellcrank, as required, to ensure push rod clearance of parking brake tube.

10) Install cotter pins and lock LH and RH push rod upper and lower attachment bolts and nuts.

11) Remove cotter pins, washers and clevis pins attaching upper end of LH and RH wheel brake rods to brake lever assembly.

12) Check for bearing wear at lower end of wheel brake rods (refer to Para 2.A.3)).

13) If evidence of wear exists, omit Para 2.A.14), and proceed as follows:

a) Remove wheel brake rods from aircraft.

b) Check that play between ball and housing does not exceed 0.020–inch, and radial play does not exceed 0.006–inch.

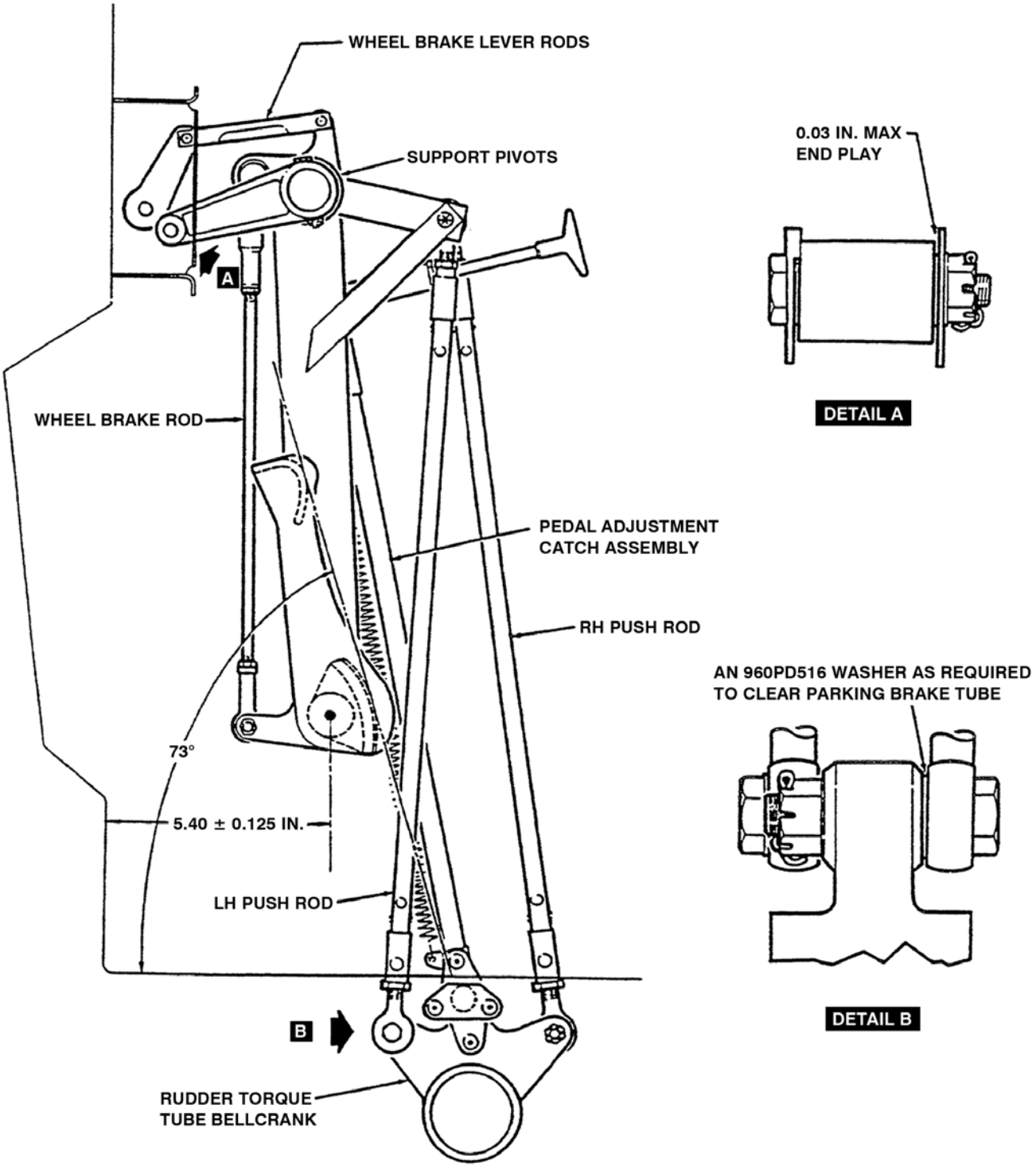
c) Check that clearance between clevis pin and wheel brake rod fork end hole does not exceed 0.018–inch.

d) Install serviceable wheel brake rods in aircraft. Connect with bolts, washers, nuts and clevis pins and lock with cotter pins.

14) If no evidence of play exists, check that clearance between clevis pin and wheel brake rod fork end hole does not exceed 0.018–inch.

15) If clearance does not exceed 0.018–inch, connect wheel brake rods to brake lever assembly, using clevis pins, washers and cotter pins.

16) Repeat Para 2.A.1) through Para 2.A.5) and check co-pilot's rudder push rods and wheel brake rod bearings.



Rudder Pedal Installation
Figure 1

B. Inspection of Rudder Torque Tube Assembly

Refer to Figure 2 for details.

Equipment required:

- Mirror
- Flashlight
- 10X Magnifying glass (or flexible borescope with 10X magnification and light)

1) Open the right-hand access panel 15, directly below cockpit (under co-pilot seat), to access the rudder torque tube and quadrant.

2) Engage the rudder quadrant gust lock by centering the rudder pedals, and lifting the gust lock lever in the floor just aft of the left (pilot) side rudder pedals.

3) Hold the rudder torque tube with one hand, then try to rotate the torque tube back and forth to check for any relative rotation (looseness) against the quadrant.

   Note There may be some small amount of free play between the gust lock plunger and the mating notch in the quadrant. This is not to be mistaken for looseness between the quadrant and torque tube.

4) Two technicians will be required to do the following inspection:

   a) While one technician pushes the rudder pedals back and forth, the other technician observes the rudder torque tube to quadrant joint for any sign of relative rotation (looseness).

   b) If any relative rotation is observed, the rudder torque tube to quadrant joint is considered loose.

5) Disengage the rudder quadrant gust lock by lowering the gust lock lever.

6) Hold the rudder quadrant with one hand and try to move the quadrant laterally (toward left-hand then right-hand direction). If any relative movement in lateral direction is observed, the rudder torque tube to quadrant joint is considered loose.

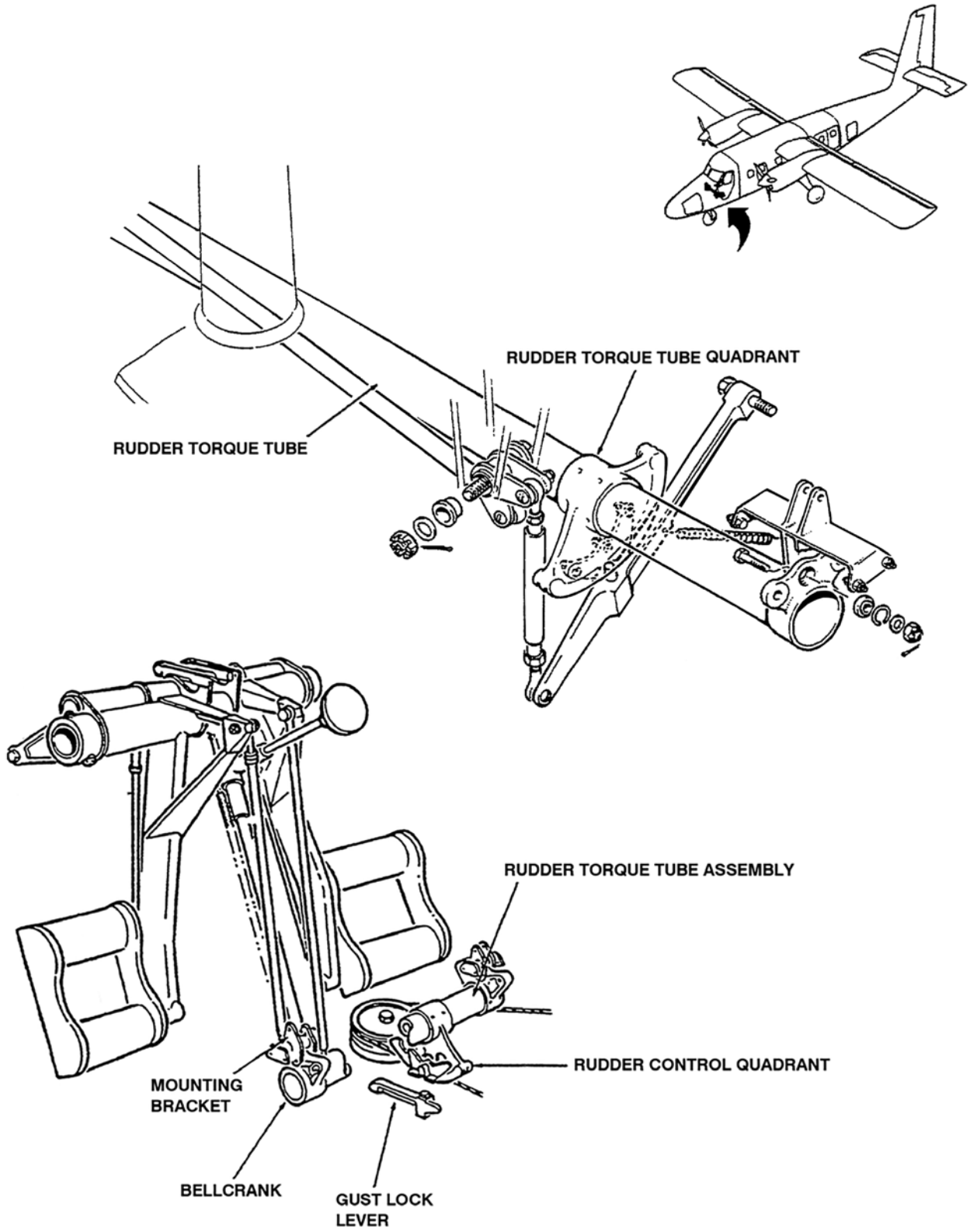
7) Use a mirror, flashlight and 10x magnification (or a flexible borescope with light) to do a detailed visual inspection of the rudder torque tube assembly for signs of loose rivets or rivet joint wear, such as dark areas or streaks around the rivet heads or quadrant to torque tube interface.

   Note There are 8 blind rivets attaching the quadrant to rudder torque tube.

8) If the rudder torque tube to quadrant joint is loose or there are signs of loose rivets or rivet joint wear, the rudder torque tube/quadrant assembly shall be replaced with a new or serviceable assembly before next flight.

9) Ensure the area is clean and clear of any swarf, tools or foreign object debris (FOD).

10) Close the access panel.



Rudder Torque Tube Assembly
Figure 2